

Introduction

Athlete performance monitoring requires reliable metrics that can accurately reflect an athlete's fatigue and injury risk. The five metrics chosen for analysis were jump height, peak propulsive force, and reactive strength index (RSI) from the Hawkins force plate tracking system, as well as accumulated acceleration load and total distance from Kinexon GPS. These metrics were selected for their strong sample sizes and relevance to understanding athlete fatigue and performance. When analyzed together over time, they can help determine whether fatigue or abnormal workload patterns can be detected longitudinally.

Jump Height

Jump height, also known as the countermovement jump (CMJ), measures how high an athlete can jump off the ground and reflects how much power they can produce in their legs. Sports technology such as force plates or timing systems measure jump height by examining flight time or ground reaction forces (Gruber et al., 2022). Jump height is important because it shows how powerful an athlete's legs are and how quickly they can produce force. Athletes who jump higher tend to sprint faster, accelerate more effectively, and change direction more explosively. Coaches also use jump height to monitor fatigue, as reductions in jump height may indicate that an athlete is overloaded (Suarez-Arrones et al., 2020). Jump height values vary by population and measurement method, with recreational male athletes typically jumping 30–40 cm, trained or collegiate athletes 40–55 cm, and elite power athletes exceeding 55 cm. Since various devices measure jump height differently, comparisons are best made within teams or using consistent testing methods (McMahon et al., 2022). Research shows that force-time variables such as take-off velocity often decline before jump height decreases, making CMJ metrics useful for identifying early signs of fatigue over time (Anicic et al., 2023). However, most research focuses

on sports such as soccer and basketball, with more research needed in collegiate women's sports. Investigating how weekly training loads affect jump height across positions could provide insights for monitoring performance and fatigue.

Peak Propulsive Force

Peak propulsive force is the maximum force produced during the propulsive phase of a jump and demonstrates an athlete's ability to generate large amounts of force in a short period of time (Bishop et al., 2021). This metric is important for rapid acceleration and has been shown to distinguish soccer players with stronger deceleration ability from those with lower deceleration capacity (Ripley et al., 2025). Peak propulsive force has been positively associated with sport-specific performance, such as clean swimming velocity in competitive swimmers (Santos et al., 2016). Reported values vary by sport and population, with professional soccer players producing approximately 28.5 N/kg for stronger athletes and 27.2 N/kg for weaker athletes (Ripley et al., 2025). Among female Olympic 100-m sprinters, peak propulsive force was reported to be 27.9 N/kg (Cabarkapa et al., 2025). Monitoring changes in peak propulsive forces has been shown to reflect improvements in strength during rehabilitation and athletic training (Hsiao et al., 2016). Similarly in athletes, tracking peak propulsive forces can show improvements in lower-body power, jumping, and sprinting. Despite its relevance, much of the literature focuses on average propulsive force rather than peak values, and there is limited research on how peak propulsive force changes over time in response to fatigue.

Reactive Strength Index (RSI)

Reactive Strength Index (RSI) measures an athlete's ability to rapidly produce force and reflects how efficiently the stretch-shortening cycle is utilized (Jarvis et al., 2022). RSI is calculated during a drop jump by dividing jump height by ground contact time (Rabelo et al., 2022). Higher

RSI values indicate that an athlete can absorb and reapply force quickly, which is essential for acceleration, agility, and change of direction performance (Rabelo et al., 2022). RSI also serves as an indicator of neuromuscular readiness, as lower values may reflect fatigue. Studies have shown that RSI demonstrates good test-retest reliability and is sensitive to fatigue (Fitzpatrick et al., 2021). RSI is important in sprinting and running, as athletes with higher RSI values tend to have shorter ground contact and more efficient strides (Southey et al., 2023). Normative values vary widely depending on sport and testing protocol, with reported means of 1.82 ± 0.44 in male volleyball players (Pleša et al., 2022) and values ranging from approximately 2.08 to 2.21 in professional male basketball players (Markwick et al., 2015). Existing research shows that RSI is meaningfully associated with athletic performance and can improve with plyometric training (Ramirez-Campillo et al., 2023). Gaps in literature include limited cross-sport comparisons and few studies linking RSI to longitudinal fatigue.

Accumulated Acceleration Load

Accumulated acceleration load represents the mechanical work generated from accelerations, decelerations, and impacts experienced during training or competition and is derived from tri-axial accelerometer data. This metric captures high-intensity, short-duration efforts that distance-based measures often miss and is used to quantify session intensity, cumulative stress, and recovery demands (Delves et al., 2021). Absolute values vary depending on device algorithms and sampling rates, so best practice involves normalizing values within teams and examining percentiles rather than universal thresholds (Freitas et al., 2025). Research shows that accumulated acceleration load provides valuable insight into the true physical demands of team sports, particularly those involving frequent changes of direction. However, there is limited

research linking acceleration-based load metrics with neuromuscular performance measures such as CMJ or RSI, despite the likelihood that sustained high acceleration loads contribute to fatigue. More research is needed to understand how different types of acceleration patterns can relate to injury risk and abnormal workload across positions and sports (Xiao et al., 2021).

Total Distance

Total distance measures the cumulative ground distance an athlete travels during a training session or match and is typically calculated using GPS or local positioning systems such as Kinexon (Kupperman & Hertel, 2020). It serves as a foundational external workload metric used to monitor movement volume, compare session demands, and manage weekly training loads (Benson et al., 2020). Match values vary by sport and position, with elite soccer players covering approximately 10–12 km per match, while court-based sports produce lower totals due to frequent stoppages (Mallo et al., 2015). Research shows that modern tracking systems provide valid and reliable distance measurements and that total distance varies by position, supporting its use in contextualizing workload (Blauberger et al., 2021; Kupperman et al., 2020). However, total distance does not capture movement intensity and is less sensitive to fatigue alone, so it is most effective when used alongside force-plate and acceleration-based metrics (Benson et al., 2020).

Research Question

Based on existing literature, this study addresses a gap in current research by asking: *How do force-plate and GPS metrics change over time, and can they identify fatigue or abnormal workload?*

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